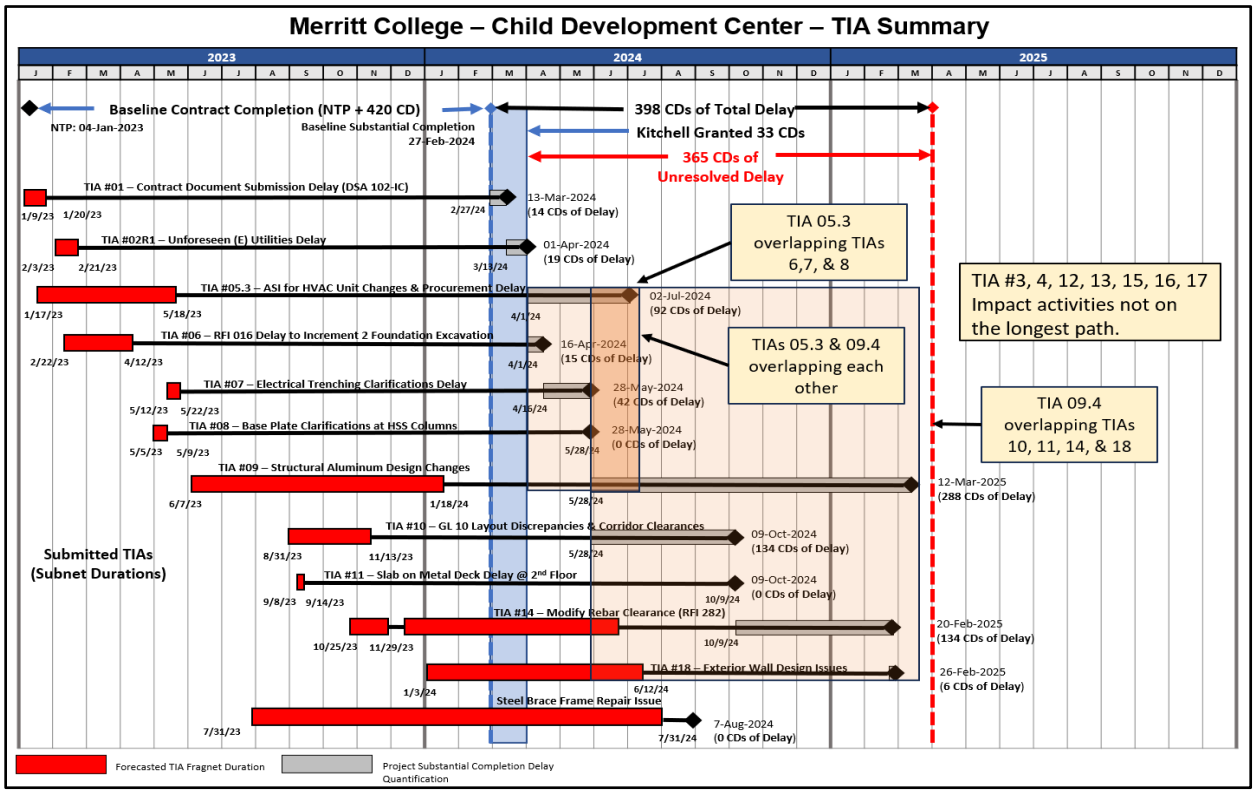
DATE:	September 03, 2024	
TO:	Bob Parks, Project Director, DBIA	Kitchell CEM
Cc:	Verna Van	Kitchell CEM
	John Armstrong, PMP, PSP, PMI-SP, CCM	Civic CM
FROM:	Jerod Bailey	Civic CM
SUBJECT:	Technical Analysis of TIA #05.3 & #09.4 Submission	
REFERENCE:	Merritt College, Child Development Center	

MEMO PURPOSE & BACKGROUND

Civic CM was retained by Kitchell CEM ("Kitchell") to provide a technical analysis of Time Impact Analyses ("TIAs") on March 11, 2024, submitted by Wickman Development & Construction ("Wickman"). Civic CM was requested to perform an independent forensic schedule analysis and to opine on Wickman's entitlement for a time extension. Civic CM elected to focus on TIAs 05.3 and 09.4 exclusively due to the concurrency of those two TIAs with all the other TIAs Wickman has submitted thus far. Wickman submitted TIA 05.3 to Kitchell on or around March 14, 2024. The original submission of TIA 05 appears to have occurred on June 19, 2023. Wickman submitted TIA 09.4 to Kitchell on or around May 20, 2024. Figure 1 below shows how TIAs 05.3 and 09.4 overlap with and with the other TIAs. A larger version of the image can be found attached to this memo in Exhibit A.

Figure 1 –TIA Summary



This memo is intended to satisfy the terms outlined in Civic CM's and Kitchell's agreement. Civic CM's analysis consisted of performing a delay trends analysis, a contemporaneous period analysis ("CPA"), a bifurcated analysis ("half step analysis"), a concurrency analysis, a technical review of the as-submitted TIAs, and an independent TIA. Civic CM also reviewed the contract documents for entitlement. The analyses were performed in accordance with appropriate portions of AAACE Recommended Practice 29R-03 Forensic Schedule Analysis, dated April 25, 2011 ("FSA 2011").

EXECUTIVE SUMMARY OF FINDINGS & RECOMMENDATIONS

Civic CM found that Wickman has not maintained entitlement for a time extension, failed to perform its TIA in accordance with the contract or published recommended practices, and its quantification of the alleged delay events is inflated. Therefore, it appears that Wickman has waived its right to a time extension and equitable adjustment to the contract for the alleged delays contemplated in TIAs 05.3 and 09.4.

Civic CM also received direction from Merritt College's program manager that resulted in a revision to its previous findings and recommendation. Civic CM found that had Wickman maintained entitlement for a time extension, it would only be entitled to a non-compensable time extension of 62 calendar days. These 62 days are attributed to two events. The first excusable delay event is the administration and execution of the ASI for the HVAC, which occurred between January through May 2023. This delay event caused 23 days of delay. The second delay event are the design revisions to the storefront. This delay event caused 39 days of delay. For both events, Civic CM identified concurrent delays that would result in an excusable but non-compensable time extension.

Civic CM also identified that Wickman is responsible for 303 days of delay. These 303 days are attributed to two events. The first non-excusable delay event is the lack of progress of structural steel erection. This delay event occurred during June through July 2023. This delay event caused 15 days of delay. The second non-delay event is Wickman failing to issue re-submittals for the storefront in a time manner. This occurred from September 2023 through June 2024. This delay event caused 288 days of delay.

BASIS OF ANALYSIS

Civic CM relied upon the following documents to satisfy the request.

1. Approved Baseline Schedule and all subsequent Monthly Schedule Updates and associated attachments and narratives through the May 2024 Schedule Update with a data date ("DD") of May 20, 2024.
2. TIA Memo - TIA Memo - "CDC_TIA 1-4_6-8_KCEM Memo.pdf"
3. TIA Memo - "CDC_TIA 1-4 Memo_2024.02.21.pdf"
4. TIA Memo - "6691 TIA 001-012 without 005009 Delay Analysis 20240115 (vv).pdf"
5. TIA Memo - "6691 TIA 001-012 without 005-009 Delay Analysis 20240122.pdf"
6. PCO Excel File - "CDC_TIA_PCO O&P_2024.0227.XLSX"

7. TIAs Native .XER File - TIAs Native .XER File - "MC-CDC BL TIA Tests-ALL.xer"
8. TIA Narrative - "Merritt CDC_TIA 1-8 Memo_2023.11.20.pdf"
9. TIA Narrative - "Merritt CDC_TIA 1-8 Memo.pdf"
10. TIA Narrative - "Merritt CDC_TIA 01-06 Delay Analysis.pdf"
11. TIA Narrative - "MERRITT CDC TIA1-8 Delay Analysis.pdf"
12. TIA Narrative - "Merritt CDC TIA #001 - DSA Form 102IC Acceptance Delay.pdf"
13. TIA Narrative - "Merritt CDC TIA #001r1 - DSA Form 102IC Acceptance Delay.pdf"
14. TIA Narrative - "Merritt CDC TIA #002 - Unforeseen Utilities Delay.pdf"
15. TIA Narrative - "Merritt CDC TIA #002r1 - Unforeseen Utilities Delay.pdf"
16. TIA Narrative - "Merritt CDC TIA #003 - Unknown (E) Sewer System - Revised Design.pdf"
17. TIA Narrative - "Merritt CDC TIA #003r1 - Unknown (E) Sewer System - Revised Design.pdf"
18. TIA Narrative - "Merritt CDC TIA #004 - Unknown Storm Drain Sewer Depth Delay.pdf"
19. TIA Narrative - "Merritt CDC TIA #005 - ASI for HVAC Unit Changes.pdf"
20. TIA Narrative - "Merritt CDC TIA #005.3 - ASI for HVAC Unit Changes Heat Pump Procurement.pdf"
21. TIA Narrative - "Merritt CDC TIA #006 - RFI 016 Delay to Increment 2 Foundation Excavation.pdf"
22. TIA Narrative - "Merritt CDC TIA #007 - Electrical Trenching Clarifications Delay.pdf"
23. TIA Narrative - "Merritt CDC TIA #008 - Base Plate Clarifications at HSS Columns.pdf"
24. TIA Narrative - "Merritt CDC TIA #09 - Structural Aluminum Design Changes.pdf"
25. TIA Narrative - "Merritt CDC TIA #09.4 - Structural Aluminum Design Changes.pdf"
26. TIA Narrative - "Merritt CDC TIA #10 - GL 10 Layout Discrepancies & Corridor Clearances DRAFT.pdf"
27. TIA Narrative - "Merritt CDC TIA #10.3 - GL 10 Layout Discrepancies & Corridor Clearances.pdf"
28. TIA Narrative - "Merritt CDC TIA #11 - Slab On Metal Deck Delay @ 2nd floor.pdf"
29. TIA Narrative - "Merritt CDC TIA #11.2 - Slab On Metal Deck Delay @ 2nd floor.pdf"
30. TIA Narrative - "Merritt CDC TIA #12 - Roof Concrete Slab Delay.pdf"
31. TIA Narrative - "Merritt CDC TIA #12.2 - Roof Concrete Slab Delay.pdf"
32. TIA Narrative - "Merritt CDC TIA #13 - Ext. Wall Framing Changes at GL E.pdf"
33. TIA Narrative - "Merritt CDC TIA #14 - Partial Roof Concrete Slab Pour (RFI 284).pdf"
34. TIA Narrative - "Merritt CDC TIA #15 - Reroute Electrical Line per ASI 005.pdf"
35. TIA Narrative - "Merritt CDC TIA #16 - Waterproofing & Slab Clarifications @ Terrace Deck.pdf"
36. TIA Narrative - "Merritt CDC TIA #17 - IBS Brace Frame Repairs.pdf"
37. TIA Narrative - "Merritt CDC TIA #18 - Exterior Framing Delay.pdf"
38. Project Manual - "01-119166_INC 02_SPC_A"
39. Notice of Delay - "NOD 01 DSA Form 102IC.pdf"
40. Notice of Delay - "NOD 02 Pad Cert. vs. RFI 016 UG Vault.pdf"
41. Notice of Delay - "NOD 03 Excessive Adverse Weather Days.pdf"
42. Notice of Delay - "NOD 04 Unknown Existing SD Depth.pdf"

43. Notice of Delay - "NOD 05 Delays to Underground Sewer Utilities.pdf"
44. Notice of Delay - "NOD 06 Waterproofing at Concrete Foundations.pdf"
45. Notice of Delay - "NOD 07 Electrical Utility Trenches and Conduits.pdf"
46. Notice of Delay - "NOD 08 Kawneer Deferred Submittal.pdf"
47. Notice of Delay - "NOD 09 RFI 200 HVAC and FP Rough In.pdf"
48. Notice of Delay - "NOD 10 RFI 228 SOMD Delay and GL 010.pdf"
49. Notice of Delay - "NOD 12 TIA 020 RFI 269 and 329, Stair 2 Delay.pdf"
50. Notice of Delay - "NOD 13 Exterior Wall Delays.pdf"
51. Notice of Delay - "NOD 14 Overdue RFI Responses.pdf"
52. Notice of Delay - "NOD 15 Overdue PCO Review.pdf"
53. Notice of Delay - "NOD 16 AE3 ASR.pdf"
54. Change Order - "NOD 16 AE3 ASR.pdf"
55. Change Order - MER CDC_PCO 023 Kawneer Custom Dies.pdf
56. Change Order - MER CDC_Wickman_CO 4_signed.pdf
57. Change Order - MER CDC_Wickman_CO 4-PCO 52.2.pdf
58. Change Order - MER CDC_Wickman_CO 06_signed.pdf
59. Change Order - MER CDC_Wickman_CO 6_PCO 29.2.pdf

ENTITLEMENT

Specification Section 00 72313 "General Conditions" Article 16.2.1 "Excusable Delay" states the following;

> *"Contractor shall, within five (5) calendar days of beginning of any delay, notify District in writing of causes of delay including documentation and facts explaining the delay and the direct correlation between the cause and effect."*

Civic CM has not received documentation to show that Wickman notified the District within five days of the delay being known. Furthermore, Article 17.7.5, "Time to Submit Proposed Change Order ("PCO")" requires the contractor to submit a PCO to the District in writing within five working days of the delay being known. It also states the following;

> *"If the contractor fails to submit the PCO within this timeframe, the contractor waives any entitlement to an adjustment of time."*

In the documentation provided to Civic CM, it does not appear that Wickman satisfied this requirement. The dates of the TIA submissions did not appear to fall within five working days of the delays being known.

Article 16.2.3 "Excusable Delay" states the following,

*"If the Contractor fails to submit justification, it waives its right to a time extension at a later date. Such **justification must be based on the official Construction Schedule as updated at the time of occurrence of the delay** or execution of work related to any changes to the Scope of Work."*

Wickman uses the baseline schedule as the basis for all 18 submitted TIAs. All the TIAs except TIA 01, 02R1, 05.3, and 06, do not use the contemporaneous schedule update at the time of the occurrence of the delay as required by the contract. Specifically, TIA 09.4 uses the Baseline schedule when the delay fragnet does not start until June 7, 2023. Wickman is not in compliance with the contract.

Article 16.2.3.2 requires that Wickman to make a good-faith effort to mitigate delays;

"…show an actual impact to the schedule, after making a good faith effort to mitigate the delay by rescheduling the work, by providing an analysis of the schedule ("Time Impact Analysis")."

Civic CM has not received any documentation or evidence to support that Wickman made a good-faith effort to mitigate the HVAC or aluminum storefront design changes related to TIA 5.3 and 9.4, respectively. Article 16.2.3.3 stipulates, "A recovery schedule must be submitted within twenty (20) calendar days of written notification to the District of causes of delay." Civic CM has not received any recovery schedule files pertaining to any of the TIAs, most importantly TIAs 5.3 and 9.4, as these are the only TIAs being analyzed.

As demonstrated in the Contract excerpts above, Civic CM cannot conclude or recommend that Wickman has satisfied the notification requirements related to delays and time extensions outlined in the Contract Documents. What is certain is that Wickman has routinely failed to prepare their TIAs in accordance with the Contract Documents, as well as industry-recognized best practices except for TIAs 01, 02R1, 05.3, and 06. TIAs 01 and 02R1 have already been granted time extensions totaling 33 CDs.

EVALUATION OF THE AS-SUBMITTED TIA

Civic CM evaluated the technical integrity of Wickman's TIAs to ensure compliance with the contract and consistency with published industry-recommended practices, such as FSA 2011. Civic CM identified that Wickman's selection of the pre-impact schedule used to host the fragnet modeling the alleged delay event for TIA 09.4 was inconsistent with FSA 2011. FSA 2011 states, "Select the as-planned network to be utilized as the un-impacted schedule. If not using the baseline, **select the contemporaneous update that existed just prior to the initial delay that is to be evaluated."**

In other words, select the contemporaneous update that existed just before the delay began to drive the critical path. Civic CM performed a concurrency analysis revealing the most critical float paths for each schedule update. This analysis showed that the revisions to the basis of design to Kawneer and other structural aluminum activities do not begin driving the critical path until the August 2023 schedule update. At the time the RFI was submitted, it was not driving the allegedly impacted work. Wickman should have selected the July 2023 schedule update to model the delay instead of the baseline schedule.

Although the selected pre-impact schedule is inconsistent with FSA 2011, the initial RFI submitted to Kitchell about this issue was transmitted on February 27, 2023. To model the impact, Wickman selected the baseline schedule with a data date of January 4, 2023. The baseline *could* be utilized but would be inappropriate given that it is inconsistent with FSA 2011 recommended practices and that the baseline is 6 months earlier than the July 2023 schedule. Wickman also submitted five schedule updates before the issue allegedly driving the critical path. Each schedule update would have more accurately shown the project's contemporaneous condition just before the start of the delay event. Performing the TIA utilizing an older schedule can make it difficult to accurately represent the effect of the delay event because an earlier schedule does not capture the project's contemporaneous conditions just prior to the start of the delay event.

Civic CM's analysis of the as-submitted TIA 05.3 shows that Wickman did not accurately model as-built progress and as-built logic changes. Wickman's fragnet does not differentiate between original scope activities or new (TIA fragnet) activities in the as-submitted TIA. Wickman used an activity code to label five activities for TIA 05.3 with only one of those activities (CDC-0448) being a new activity. The activity CDC-0448 ASI 003 & 006 For HVAC Unit Changes (ongoing TIA eliminated for TIA Tests) included an original forecast duration of 60 days. In the schedule update for May 2023, the activity was marked complete with an actual start on April 28, 2023 and an actual finish on May 18, 2023 for an actual duration of 15 days. This led to an over forecast in the TIA of 45 days. In evaluating the timelines from both Wickman and Kitchell, we see significant activities occurring during this period and some significant gaps in the timeline. There are six submittal activities, seven revise and resubmit activities, and three approval activities that should have been included in the fragnet for TIA 05.3, instead of only one hammock activity for 60 days. After putting all these activities in chronological order, two significant gaps stand out:

- Gap of 41 days after AOR review of Submittal-070.1 and before Wickman's submission of Submittal-175
- Gap of 23 days after approval to release heat pump for procurement and before WDC issues PCO-029.1 for costs related to ASI-003 changes. CalAir (HVAC subcontractor) requested a new CCD for the revised heat pump before they purchased it.

Totaling all these gaps in the timeline for TIA 05.3 yields a 64-day portion of the delay overshadowed by the hammock activity CDC-0448. More importantly, this portion of the delay timeline is when the ball is in Wickman's court.

Civic CM's analysis of the as-submitted TIA 09.4 shows that Wickman did not accurately model as-built progress and as-built logic changes. Wickman's fragnet does not differentiate between original scope activities or new (TIA fragnet) activities in the as-submitted TIA. Wickman used an activity code to label 4 activities for TIA 09.4 with only one (CDC-0642) being a new activity. The activity CDC-0642 Revisions to Basis of Design to Kawneer is a lump sum duration activity intended to demonstrate the entire 214-day delay. In evaluating the timelines from Wickman and Kitchell, significant activities are occurring during this period, and there are some significant gaps in the timeline. There are 14 submittal activities, 16 revise and resubmit activities, 4 approval activities, 7 RFI submissions activities, and 7 RFI response activities that should have been included in the fragnet for TIA 09.4, instead of only one lump sum duration activity. After putting all these activities in chronological order, several significant gaps stand out:

- Gap of 24 days after AOR review of Submittal-095 and before Wickman's submission of RFI-033
- Gap of 42 days after AOR review of RFI-040 and before Wickman's Submittal-095.2
- Gap of 5 days after AOR review of Submittal-095.2 and before Wickman's submission of RFI-135
- Gap of 10 days after AOR review of RFI-135 and before Wickman's submission of RFI-151
- Gap of 11 days after AOR review of RFI-169 and before Wickman's submission of RFI-179
- Gap of 4 days after AOR review of Submittal-095.5 and Wickman's preparation activity for Submittal-095.6
- Gap of 7 days after AOR review of Submittal-095.6 and Wickman's second submission of Submittal-095.6
- Gap of 18 days after AOR review of Submittal-095.7 and Wickman's Submittal-095.8

Totaling all these gaps in the timeline for TIA 09.4 yields 121 days of delay that was overshadowed by the lump sum activity CDC-0642. More importantly, this portion of the delay is when Wickman is responsible for the next action. In other words, Wickman owns 121 days of this 365-day delay.

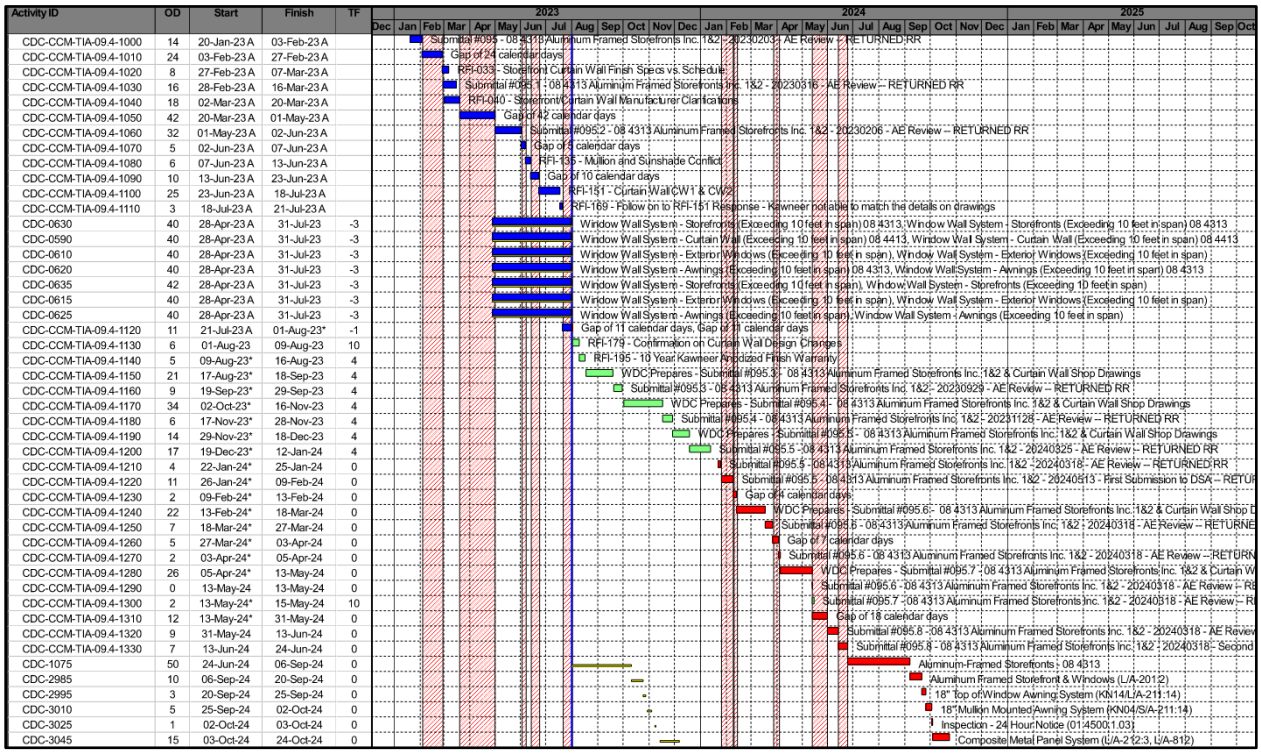
RESULTS OF THE INDEPENDENTLY PREPARED TIA

Civic CM's analysis of TIA 09.4 included inserting the timeline of events, including the gaps referenced above, into the proper pre-impact schedule update file of July 2023. This accounts for all the changes that have been realized to date (31-Jul-2023), giving the most accurate picture of the as-built condition at the time just before the impact. After making said changes to the impacted work, the new forecasted Substantial Completion date (CDC-0220) is March 24, 2025, or a 15 CD improvement to the Substantial Completion date compared to what Wickman reported in its TIA

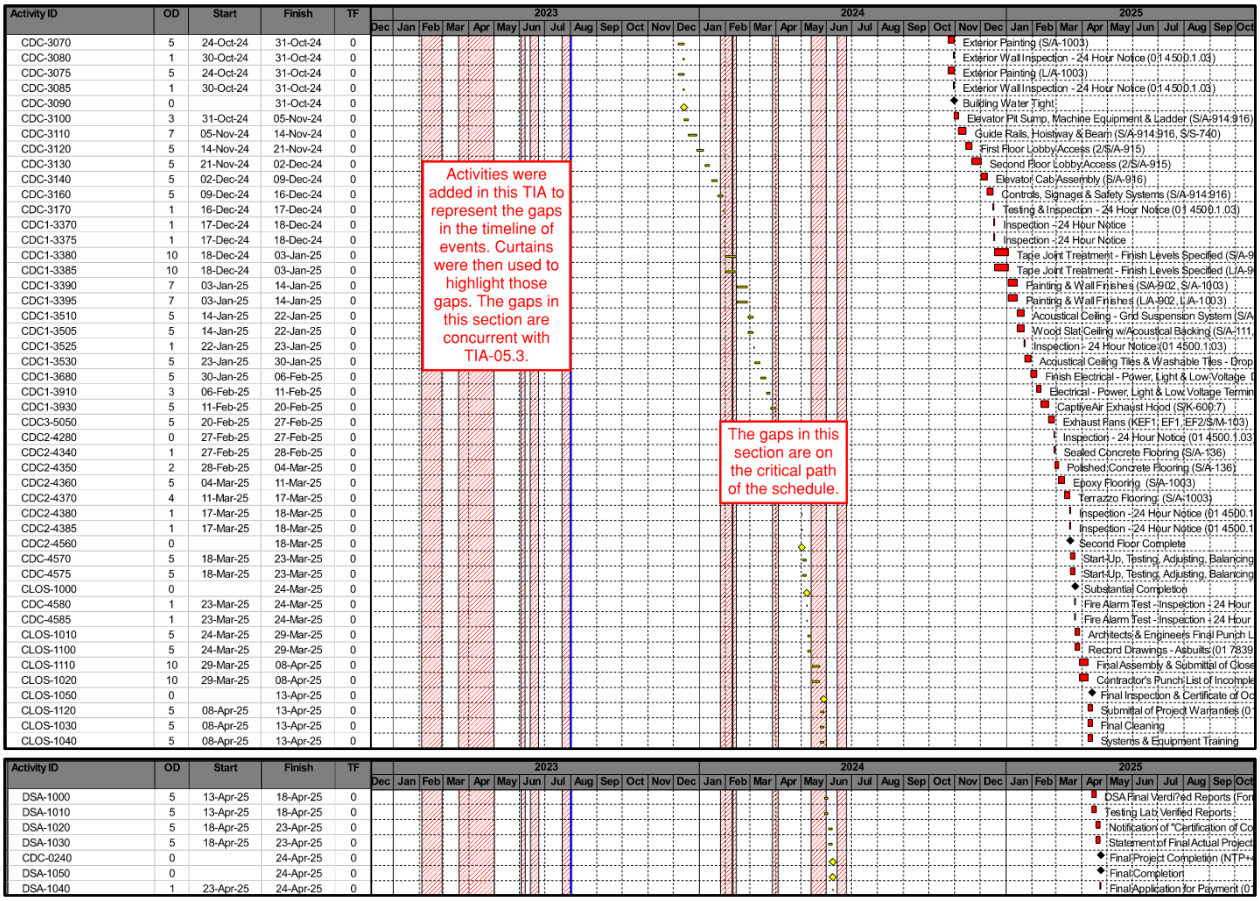
submission (31-Mar-2025). March 24, 2025, amounts to 358 days past the current approved Substantial Completion date of March 31, 2024. Without additional information regarding what work was taking place during the referenced time gaps, Civic CM is left to assume that the work would have been completed 121 days earlier than what is shown in the as-submitted TIA.

The image below is of the independent TIA performed by Civic CM. The timeline gaps were included as activities in this TIA, and curtains were used over those activities and the DSA approval activities to demonstrate their significance. The below is included in Exhibit B with this memo.

Figure 2 – CCM Independent TIA Including Timeline of Events on July 2023 Update



Activity ID	OD	Start	Finish	TF
CDC-CCM-TIA-09.4-1000	14	20-Jan-23 A	03-Feb-23 A	
CDC-CCM-TIA-09.4-1010	24	03-Feb-23 A	27-Feb-23 A	
CDC-CCM-TIA-09.4-1020	8	27-Feb-23 A	07-Mar-23 A	
CDC-CCM-TIA-09.4-1030	16	28-Feb-23 A	16-Mar-23 A	
CDC-CCM-TIA-09.4-1040	18	02-Mar-23 A	20-Mar-23 A	
CDC-CCM-TIA-09.4-1050	42	20-Mar-23 A	01-May-23 A	
CDC-CCM-TIA-09.4-1060	32	01-May-23 A	02-Jun-23 A	
CDC-CCM-TIA-09.4-1070	5	02-Jun-23 A	07-Jun-23 A	
CDC-CCM-TIA-09.4-1080	6	07-Jun-23 A	13-Jun-23 A	
CDC-CCM-TIA-09.4-1090	10	13-Jun-23 A	23-Jun-23 A	
CDC-CCM-TIA-09.4-1100	25	23-Jun-23 A	18-Jul-23 A	
CDC-CCM-TIA-09.4-1110	3	18-Jul-23 A	21-Jul-23 A	
CDC-0630	40	28-Apr-23 A	31-Jul-23	-3
CDC-0590	40	28-Apr-23 A	31-Jul-23	-3
CDC-0610	40	28-Apr-23 A	31-Jul-23	-3
CDC-0620	40	28-Apr-23 A	31-Jul-23	-3
CDC-0635	42	28-Apr-23 A	31-Jul-23	-3
CDC-0615	40	28-Apr-23 A	31-Jul-23	-3
CDC-0625	40	28-Apr-23 A	31-Jul-23	-3
CDC-CCM-TIA-09.4-1120	11	21-Jul-23 A	01-Aug-23*	-1
CDC-CCM-TIA-09.4-1130	6	01-Aug-23	09-Aug-23	10
CDC-CCM-TIA-09.4-1140	5	09-Aug-23*	16-Aug-23	4
CDC-CCM-TIA-09.4-1150	21	17-Aug-23*	18-Sep-23	4
CDC-CCM-TIA-09.4-1160	9	19-Sep-23*	29-Sep-23	4
CDC-CCM-TIA-09.4-1170	34	02-Oct-23*	16-Nov-23	4
CDC-CCM-TIA-09.4-1180	6	17-Nov-23*	28-Nov-23	4
CDC-CCM-TIA-09.4-1190	14	29-Nov-23*	18-Dec-23	4
CDC-CCM-TIA-09.4-1200	17	19-Dec-23*	12-Jan-24	4
CDC-CCM-TIA-09.4-1210	4	22-Jan-24*	25-Jan-24	0
CDC-CCM-TIA-09.4-1220	11	26-Jan-24*	09-Feb-24	0
CDC-CCM-TIA-09.4-1230	2	09-Feb-24*	13-Feb-24	0
CDC-CCM-TIA-09.4-1240	22	13-Feb-24*	18-Mar-24	0
CDC-CCM-TIA-09.4-1250	7	18-Mar-24*	27-Mar-24	0
CDC-CCM-TIA-09.4-1260	5	27-Mar-24*	03-Apr-24	0
CDC-CCM-TIA-09.4-1270	2	03-Apr-24*	05-Apr-24	0
CDC-CCM-TIA-09.4-1280	26	05-Apr-24*	13-May-24	0
CDC-CCM-TIA-09.4-1290	0	13-May-24	13-May-24	0
CDC-CCM-TIA-09.4-1300	2	13-May-24*	15-May-24	10
CDC-CCM-TIA-09.4-1310	12	13-May-24*	31-May-24	0
CDC-CCM-TIA-09.4-1320	9	31-May-24	13-Jun-24	0
CDC-CCM-TIA-09.4-1330	7	13-Jun-24	24-Jun-24	0
CDC-1075	50	24-Jun-24	06-Sep-24	0
CDC-2985	10	06-Sep-24	20-Sep-24	0
CDC-2995	3	20-Sep-24	25-Sep-24	0
CDC-3010	5	25-Sep-24	02-Oct-24	0
CDC-3025	1	02-Oct-24	03-Oct-24	0
CDC-3045	15	03-Oct-24	24-Oct-24	0



CONTEMPORANEOUS PERIOD ANALYSIS

A contemporaneous period analysis ("CPA") analyzes delays to a project or a specific completion milestone. A CPA periodically analyzes delays or gains to selected milestones and identifies the driving Critical Path activities for the selected milestone(s) (typically between updates). Classified as retrospective, this type of analysis represents delays as they are incurred contemporaneously accurately. It is generally an excellent way to get a high-level understanding of past issues on a Project, given that the schedule itself is appropriately managed. CPAs are typically utilized as an initial analysis for projects that have experienced schedule delays. The CPA allows the analyst to review what schedule activities drove the project on a period-by-period basis and gives an initial history of the schedule's journey throughout the project's duration. CPAs often identify specific areas which merit further investigation and analysis. Additionally, the CPA can analyze other schedule-related performance indicators, such as the potential for concurrent delay.

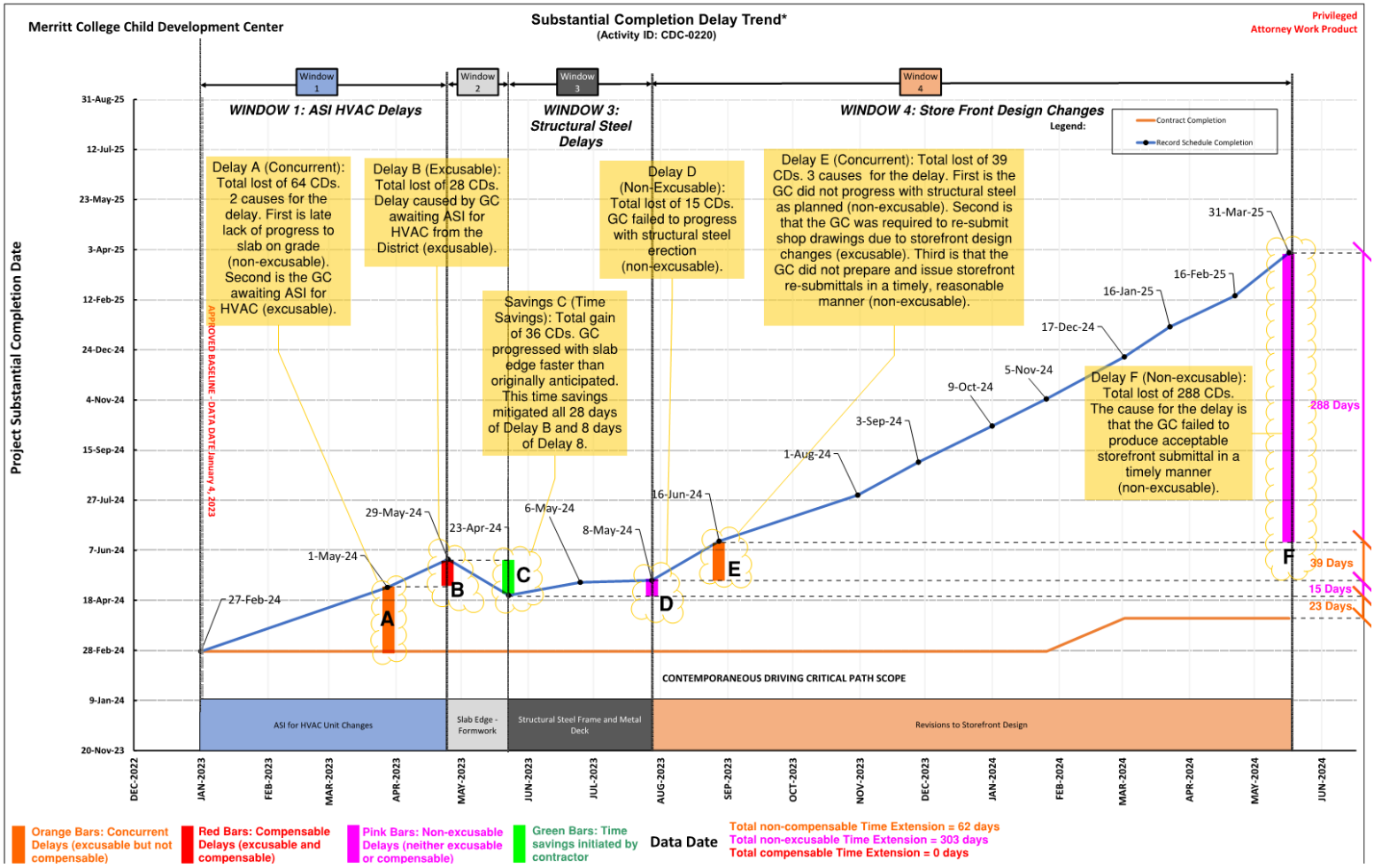
Furthermore, the Bifurcated Analysis, also known as a Half-Step Analysis, is effectively an enhanced version of the CPA that allows the analyst to separate impacts or improvements to a schedule's Completion date caused by different factors, such as progress delays or improvements, logic modifications, or other schedule modifications. The process is such that the analyst will separate updates to update schedule changes using solely pure progress from the

current schedule update and record what the Completion date of the milestone would have been if not for additional schedule modifications.

Civic CM performed the CPA and Bifurcated Analysis in accordance with FSA 2011 MIPs 3.3 and 3.4. The image below shows the graphical representation of the CPA results and associated components. The CPA is represented by the orange line (the current Contract Completion Date), the blue line (Forecasted Substantial Completion Date), and the Contemporaneous Driving Critical Path at the bottom, which identifies the specific activity from the schedule that was driving the Substantial Completion date for the schedule update with the corresponding data date. For example, in the April 2023 Schedule Update (DD: 28-Apr-23), activity SP.050-100 – "Forthcoming ASI for HVAC Unit Changes (Place Holder)" was the driving activity causing the Substantial Completion to be April 23, 2024.

The Bifurcated Analysis is represented by the Black Line (pure progress results from performing the Bifurcated Analysis), vertical green dotted lines (mitigation resulting from schedule modifications, not including progress updates), and the vertical red dotted lines (additional delay caused by schedule modifications, not including progress updates). Impacts to Substantial Completion are represented by the vertical red bars and callouts.

Figure 3 – Contemporaneous Period Analysis



A larger version of the graphic above can be found in Exhibit C, attached to this memo.

Window 1: Spans from January 4, 2023, to April 28, 2023

At the start of the window, the schedule forecasted a completion date of February 27, 2024. The schedule forecasted a completion date of May 29, 2024, at the end of the window. This is a slippage of 92 CDs.

The first window is driven by the ASI for HVAC Unit Changes activity. It is the reason for the 92-day delay in Substantial Completion from February 27, 2024, to May 29, 2024, during the first window. Also occurring during this window were critical slippages to pouring the slab-on-grade (activity CDC-0160). During the first period, this activity did not progress as previously anticipated, and due to the lack of progress, the substantial completion date was pushed to May 01, 2024. This was identified in the bifurcated analysis.

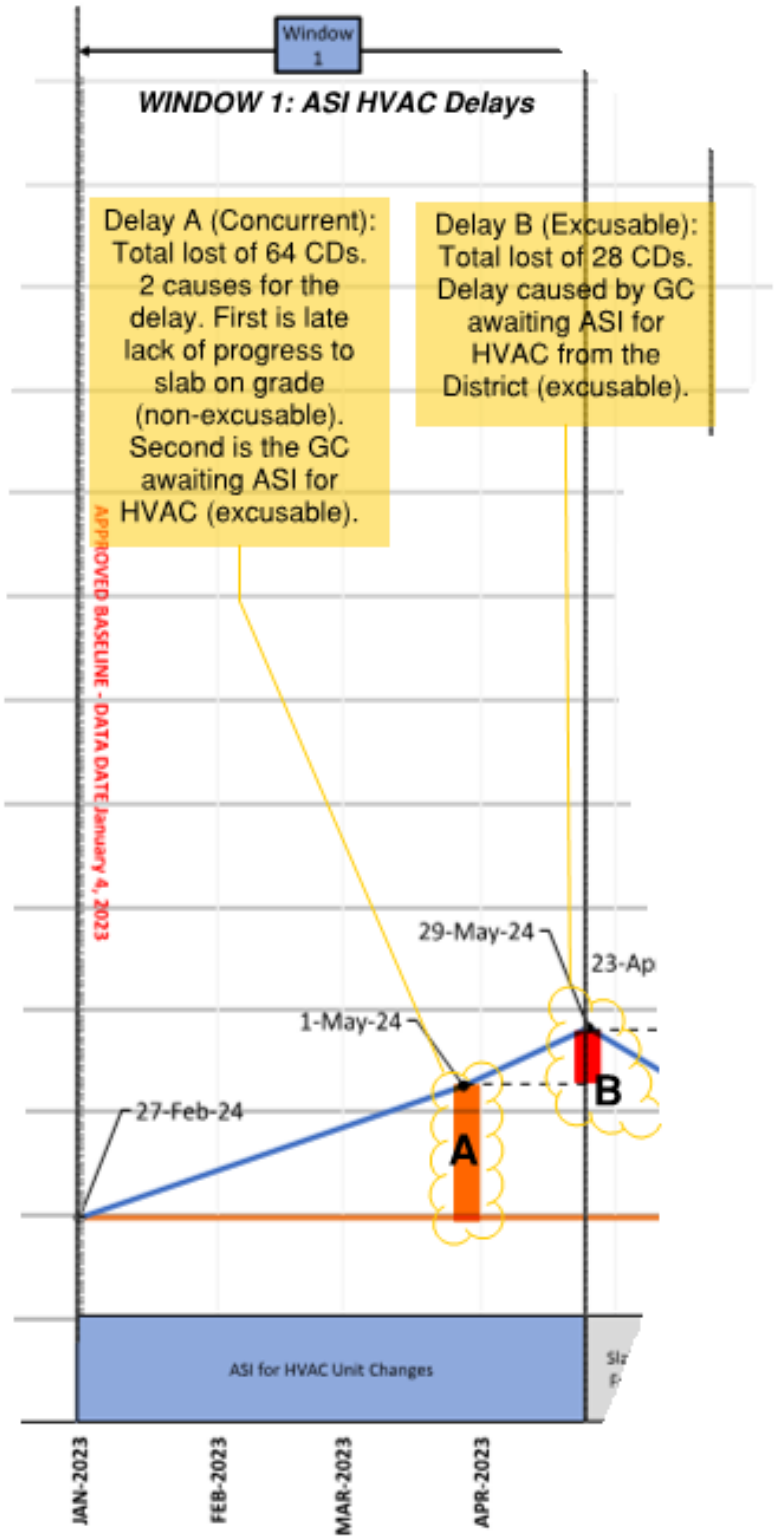
Table 1 - Window 1 Driving Activities

Schedule File	Schedule Update	Data Date	Record Schedule Completion	Update Delay	Cumulative Delay	Half Step	Mitigation	Cumulative Mitigation	CP Driving Activity ID	Critical Path Driving Activity Name
24-50_2023.01_BL	Jan-23	1/4/2023	27-Feb-24	0	0	27-Feb-24			CDC-0100	Notice to Proceed (005500)
	Mar 2023 Half Step	3/31/2023				1-May-24	0	0	CDC-0160	Slab-On-Grade
24-50_2023.03	Mar-23	3/31/2023	1-May-24	-64	-64	1-May-24			CDC-0448	Forcoming ASI for HVAC Unit Changes (Place Holder)
	Apr 2023 Half Step	4/28/2023				29-May-24	0	0	CDC-0448	Forcoming ASI for HVAC Unit Changes (Place Holder)
24-50_2023.04	Apr-23	4/28/2023	29-May-24	-28	-92	29-May-24			CDC-0448	Forcoming ASI for HVAC Unit Changes (Place Holder)

If the March 2023 schedule update had no activity or logic changes, activity CDC-0160 would have been the driving activity of the critical path. Changes made in the March 2023 schedule update include changing the activity type of CDC-0160 from task-dependent to level of effort and adding relationships to the activity CDC-0448, the ASI for HVAC Unit Changes activity.

Civic CM's analysis indicates a concurrent delay within the first update period. Referencing Delay A[1] in Exhibit C, the lack of progress for the slab was occurring at the same time as the ASI delay and had equal impact on the critical path. Referencing Delay B[2] in Exhibit C, Civic CM did not identify a concurrent delay.

[1] Delay A (Exhibit C) = Total 64 days of delay, concurrent
[2] Delay B (Exhibit C) = Total 28 days of delay, no concurrency


Window 1
WINDOW 1: ASI HVAC Delays
Delay A (Concurrent): Total lost of 64 CDs. 2 causes for the delay. First is late lack of progress to slab on grade (non-excusable). Second is the GC awaiting ASI for HVAC (excusable).
Delay B (Excusable): Total lost of 28 CDs. Delay caused by GC awaiting ASI for HVAC from the District (excusable).
APPROVED BASELINE - DATA DATE January 4, 2023
29-May-24
23-Ap
1-May-24
27-Feb-24
A
B
ASI for HVAC Unit Changes
JAN-2023
FEB-2023
MAR-2023
APR-2023

Window 2: Spans from April 28, 2023 to May 26, 2023

At the start of the window, the schedule forecasted a completion date of May 29, 2024. The schedule forecasted a completion date of April 23, 2024, at the end of the window. This is a savings of 36 CDs.

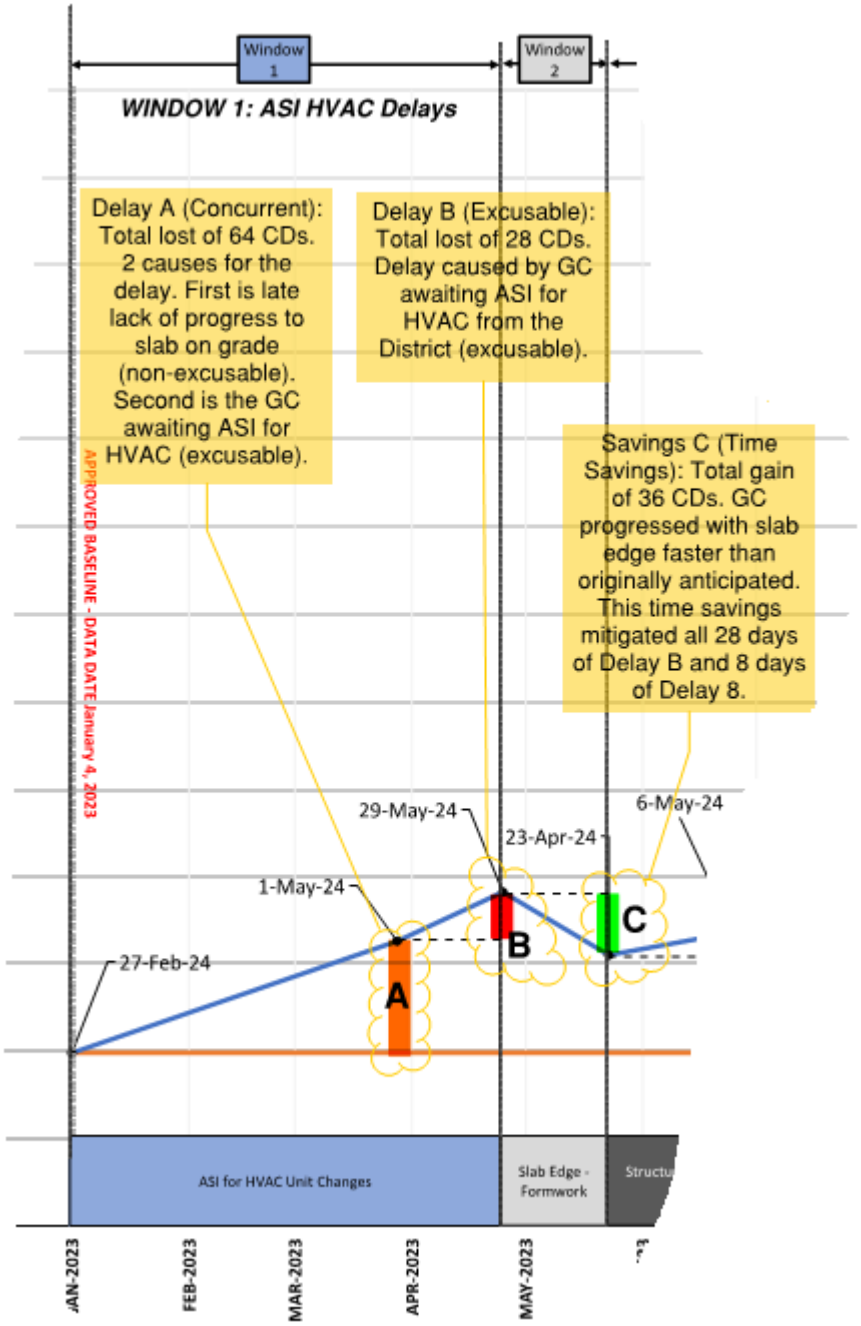
The critical path during this period is the slab edge forming (activity CDC1-2170). Wickman's progress with this scope appears to have mitigated 36 CDs of delay experienced in Window 1. This window also does not appear to be the subject of any TIAs, let alone the two TIAs in question.

Table 2 - Window 2 Driving Activities

Schedule File	Schedule Update	Data Date	Record Schedule Completion	Update Delay	Cumulative Delay	Half Step	Mitigation	Cumulative Mitigation	CP Driving Activity ID	Critical Path Driving Activity Name
24-50_2023.04	Apr-23	4/28/2023	29-May-24	-28	-92	29-May-24			CDC-0448	Forcoming ASI for HVAC Unit Changes (Place Holder)
	May 2023 Half Step	5/26/2023				23-Apr-24	0	0	CDC1-2170	Slab Edge - Formwork, Blockouts, & Depressions (S/S-101)
24-50_2023.05	May-23	5/26/2023	23-Apr-24	36	-56	23-Apr-24			CDC1-2170	Slab Edge - Formwork, Blockouts, & Depressions (S/S-101)

Referencing Savings C in Exhibit C, Civic CM's analysis found that the 'better than planned' progress for the slab edge form saved 36 days. This savings mitigated all 28 days associated with Delay B and eight days of Delay A. And accounting for previously executed time extension totaling 33 days, it appears Wickman is only entitled[3] to a time extension of 23 days[4].

[3] This assumes Wickman provided noticed and met other requirements of the prime contract
[4] 23 days = Delay A days + Delay B days + Savings C days – previously executed time extensions = 64 + 28 -15 – 33



Window 3: Spans from May 26, 2023 to July 31, 2023

At the start of the window, the schedule forecasted a completion date of April 23, 2024. The schedule forecasted a completion date of May 8, 2024, at the end of the window. This is a slippage of 15 CDs.

Table 3 - Window 3 Driving Activities

Schedule File	Schedule Update	Data Date	Record Schedule Completion	Update Delay	Cumulative Delay	Half Step	Mitigation	Cumulative Mitigation	CP Driving Activity ID	Critical Path Driving Activity Name
24-50_2023.05	May-23	5/26/2023	23-Apr-24	36	-56	23-Apr-24			CDC1-2170	Slab Edge - Formwork, Blockouts, & Depressions (S/S-101)
	Jun 2023 Half Step	6/28/2023				12-May-24	6	6	CDC1-2120	Testing & Inspection - 24-Hrs Prior to Backfill (01 4500.1.03)
24-50_2023.06	Jun-23	6/28/2023	6-May-24	-13	-69	6-May-24			CDC-2405	Structural Steel Frame - Columns, Braceframes, Girders & Beams
	Jul 2023 Half Step	7/31/2023				12-May-24	4	10	CDC-2520	Stair-03 - Stringer Stairs & Connections - Roof (S/A-704:05, S/S-104, S-730)
24-50_2023.07	Jul-23	7/31/2023	8-May-24	-2	-71	8-May-24			CDC2-2600	Metal Deck - Second Floor (S/S-102, S-710)

The critical driving force through the as-submitted schedules is structural steel erection and metal decking. The other submitted TIAs do not appear to account for these particular delays. The time lost during this period appears non-excusable, given the information provided to Civic CM to date.

Window 4: Spans from July 31, 2023 to May 20, 2024

At the start of the window, the schedule forecasted a completion date of May 8, 2024. The schedule forecasted a completion date of March 31, 2025, at the end of the window. This is a slippage of 327 CDs.

Table 4 - Window 4 Driving Activities

Schedule File	Schedule Update	Data Date	Record Schedule Completion	Update Delay	Cumulative Delay	Half Step	Mitigation	Cumulative Mitigation	CP Driving Activity ID	Critical Path Driving Activity Name
24-50_2023.07	Jul-23	7/31/2023	8-May-24	-2	-71	8-May-24			CDC2-2600	Metal Deck - Second Floor (S/S-102, S-710)
	Aug 2023 Half Step	8/31/2023				11-Jun-24	-5	5	CDC2-2605	Metal Deck - Second Floor (S/L-102, L-710)
24-50_2023.08	Aug-23	8/31/2023	16-Jun-24	-39	-110	16-Jun-24			CDC-0642	Revisions to Basis of Design to Kawneer
	Oct 2023 Half Step	11/3/2023				1-Aug-24	0	5	CDC-1075	Aluminum-Framed Storefronts - 08 4313
24-50_2023.10	Oct-23	11/3/2023	1-Aug-24	-46	-156	1-Aug-24			CDC-1075	Aluminum-Framed Storefronts - 08 4313
	Nov 2023 Half Step	12/1/2023				30-Sep-24	27	32	CDC-0642	Revisions to Basis of Design to Kawneer
24-50_2023.11	Nov-23	12/1/2023	3-Sep-24	-33	-189	3-Sep-24			CDC-2802	Curb Layout and Adjustments at GL 10 (RFI 178 and 180)
	Dec 2023 Half Step	1/4/2024				30-Sep-24	-9	23	CDC-0642	Revisions to Basis of Design to Kawneer
24-50_2023.12	Dec-23	1/4/2024	9-Oct-24	-36	-225	9-Oct-24			CDC-0642	Revisions to Basis of Design to Kawneer
	Jan 2024 Half Step	1/29/2024				22-Oct-24	-14	9	CDC-0642	Revisions to Basis of Design to Kawneer
24-50_2024.01	Jan-24	1/29/2024	5-Nov-24	-27	-252	5-Nov-24			CDC-0642	Revisions to Basis of Design to Kawneer
	Feb 2024 Half Step	3/5/2024				25-Nov-24	-22	-13	CDC-0642	Revisions to Basis of Design to Kawneer
24-50_2024.02	Feb-24	3/5/2024	17-Dec-24	-42	-261	17-Dec-24			CDC-0642	Revisions to Basis of Design to Kawneer
	Mar 2024 Half Step	3/26/2024				18-Dec-24	-29	-42	CDC-0642	Revisions to Basis of Design to Kawneer
24-50_2024.03	Mar-24	3/26/2024	16-Jan-25	-30	-291	16-Jan-25			CDC-0642	Revisions to Basis of Design to Kawneer
	Apr 2024 Half Step	4/25/2024				21-Jan-25	-26	-68	CDC-0642	Revisions to Basis of Design to Kawneer
24-50_2024.04	Apr-24	4/25/2024	16-Feb-25	-31	-322	16-Feb-25			CDC-0642	Revisions to Basis of Design to Kawneer
	May 2024 Half Step	5/20/2024				16-Feb-25	-43	-111	CDC-1075	Aluminum-Framed Storefronts - 08 4313
24-50_2024.05	May-24	5/20/2024	31-Mar-25	-43	-365	31-Mar-25			CDC-0642	Revisions to Basis of Design to Kawneer

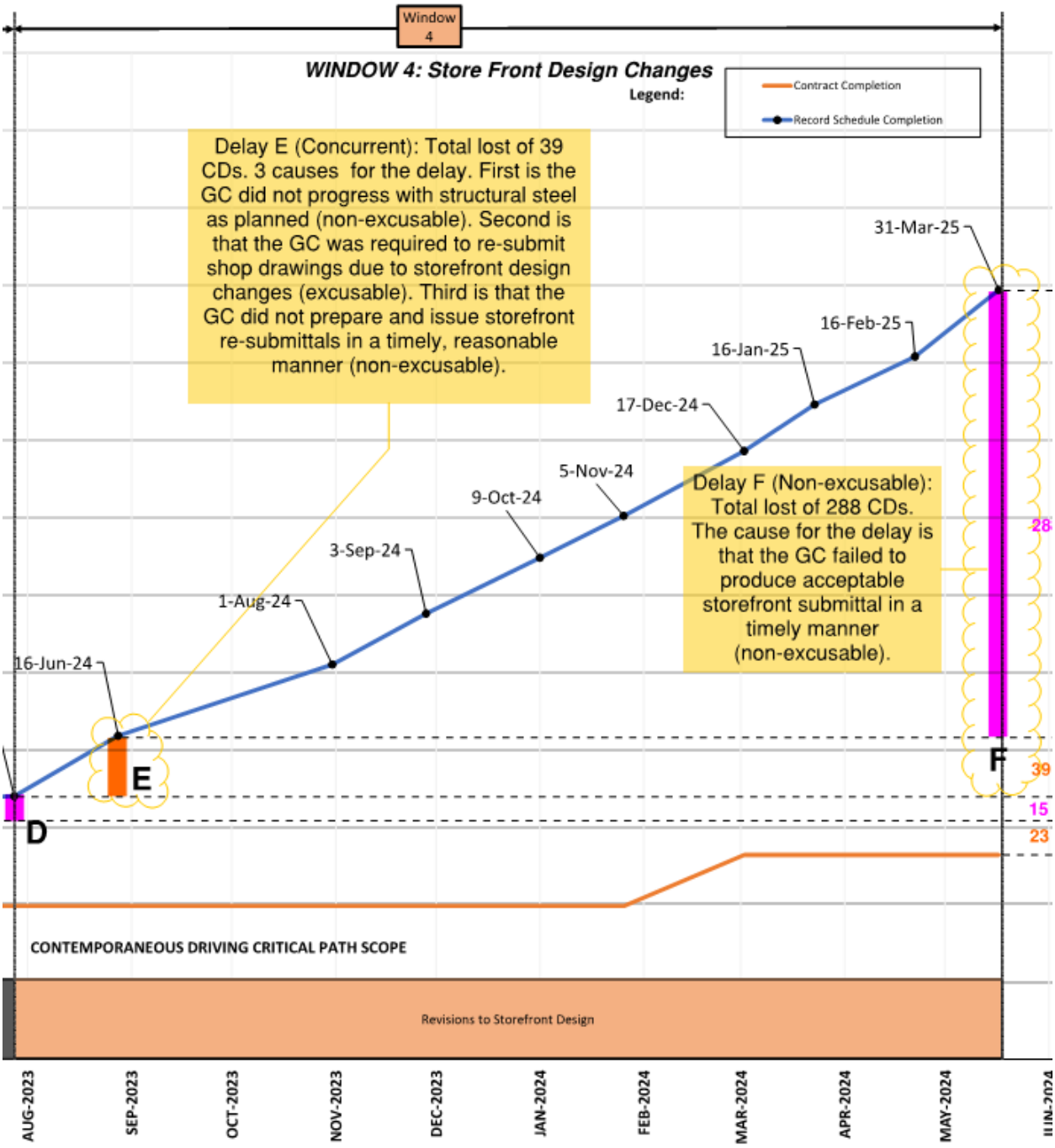
The preponderance of the driving critical path activities during this window appears to be related to the alleged delay events associated with TIA 09.4. However, at the start of the window, delays associated with the lack of meeting planned progress for the metal decking scope continue to drive the critical path. This is demonstrated in Civic CM's bifurcated analysis. Absent the logic and activity changes made by Wickman in its August 2023 schedule update, the substantial completion date would have slipped 34 CDs.

Also occurring in the first update period, Wickman was preparing revised submissions for the design changes. This occurred in August 2023. As previously mentioned, Civic CM identified periods in which Wickman did not progress certain work elements associated with TIA 09.4 promptly and judiciously. Below is a timeline of events associated with TIA 09.4. Upon evaluation of the timeline, it is readily apparent that work is not ongoing continuously.

Figure 4 – TIA 09.4 Timeline of Events

TIA 09.4 Timeline			
Activity ID	P6 Activity Name	Start	Finish
CDC-CCM-TIA-1000	Submittal #095 - 08 4313 Aluminum Framed Storefronts Inc. 1&2 - 20230203 - AE Review -- RETURNED RR	1/20/2023	2/3/2023
CDC-CCM-TIA-1010	Gap of 24 calendar days	2/3/2023	2/27/2023
CDC-CCM-TIA-1020	RFI-033 - Storefront Curtain Wall Finish Specs vs. Schedule	2/27/2023	3/7/2023
CDC-CCM-TIA-1030	Submittal #095.1 - 08 4313 Aluminum Framed Storefronts Inc. 1&2 - 20230316 - AE Review -- RETURNED RR	2/28/2023	3/16/2023
CDC-CCM-TIA-1040	RFI-040 - Storefront/Curtain Wall Manufacturer Clarifications	3/2/2023	3/20/2023
CDC-CCM-TIA-1050	Gap of 42 calendar days	3/20/2023	5/1/2023
CDC-CCM-TIA-1060	Submittal #095.2 - 08 4313 Aluminum Framed Storefronts Inc. 1&2 - 20230206 - AE Review -- RETURNED RR	5/1/2023	6/2/2023
CDC-CCM-TIA-1070	Gap of 5 calendar days	6/2/2023	6/7/2023
CDC-CCM-TIA-1080	RFI-135 - Mullion and Sunshade Conflict	6/7/2023	6/13/2023
CDC-CCM-TIA-1090	Gap of 10 calendar days	6/13/2023	6/23/2023
CDC-CCM-TIA-1100	RFI-151 - Curtain Wall CW1 & CW2	6/23/2023	7/18/2023
CDC-CCM-TIA-1110	RFI-169 - Follow on to RFI-151 Response - Kawneer not able to match the details on drawings	7/18/2023	7/21/2023
CDC-CCM-TIA-1120	Gap of 11 calendar days	7/21/2023	8/1/2023
CDC-CCM-TIA-1130	RFI-179 - Confirmation on Curtain Wall Design Changes	8/1/2023	8/9/2023
CDC-CCM-TIA-1140	RFI-195 - 10 Year Kawneer Anodized Finish Warranty	8/9/2023	8/16/2023
CDC-CCM-TIA-1150	WDC Prepares - Submittal #095.3 - 08 4313 Aluminum Framed Storefronts Inc. 1&2 & Curtain Wall Shop Drawings	8/17/2023	9/18/2023
CDC-CCM-TIA-1160	Submittal #095.3 - 08 4313 Aluminum Framed Storefronts Inc. 1&2 - 20230929 - AE Review -- RETURNED RR	9/19/2023	9/29/2023
CDC-CCM-TIA-1170	WDC Prepares - Submittal #095.4 - 08 4313 Aluminum Framed Storefronts Inc. 1&2 & Curtain Wall Shop Drawings	9/30/2023	11/16/2023
CDC-CCM-TIA-1180	Submittal #095.4 - 08 4313 Aluminum Framed Storefronts Inc. 1&2 - 20231128 - AE Review -- RETURNED RR	11/17/2023	11/28/2023
CDC-CCM-TIA-1190	WDC Prepares - Submittal #095.5 - 08 4313 Aluminum Framed Storefronts Inc. 1&2 & Curtain Wall Shop Drawings	11/29/2023	12/18/2023
CDC-CCM-TIA-1200	Submittal #095.5 - 08 4313 Aluminum Framed Storefronts Inc. 1&2 - 20240325 - AE Review -- RETURNED RR	12/19/2023	1/12/2024
CDC-CCM-TIA-1210	Submittal #095.5 - 08 4313 Aluminum Framed Storefronts Inc. 1&2 - 20240318 - AE Review -- RETURNED RR	1/22/2024	1/25/2024
CDC-CCM-TIA-1220	Submittal #095.5 - 08 4313 Aluminum Framed Storefronts Inc. 1&2 - 20240513 - First Submission to DSA -- RETURNED RR	1/25/2024	2/9/2024
CDC-CCM-TIA-1230	Gap of 4 calendar days	2/9/2024	2/13/2024
CDC-CCM-TIA-1240	WDC Prepares - Submittal #095.6 - 08 4313 Aluminum Framed Storefronts Inc. 1&2 & Curtain Wall Shop Drawings	2/13/2024	3/17/2024
CDC-CCM-TIA-1250	Submittal #095.6 - 08 4313 Aluminum Framed Storefronts Inc. 1&2 - 20240318 - AE Review -- RETURNED RR	3/18/2024	3/27/2024
CDC-CCM-TIA-1260	Gap of 7 calendar days	3/27/2024	4/3/2024
CDC-CCM-TIA-1270	Submittal #095.6 - 08 4313 Aluminum Framed Storefronts Inc. 1&2 - 20240318 - AE Review -- RETURNED RR	4/3/2024	4/5/2024
CDC-CCM-TIA-1280	WDC Prepares - Submittal #095.7 - 08 4313 Aluminum Framed Storefronts Inc. 1&2 & Curtain Wall Shop Drawings	4/5/2024	5/12/2024
CDC-CCM-TIA-1300	Submittal #095.6 - 08 4313 Aluminum Framed Storefronts Inc. 1&2 - 20240318 - AE Review -- RETURNED RR	5/13/2024	5/15/2024
CDC-CCM-TIA-1290	Submittal #095.7 - 08 4313 Aluminum Framed Storefronts Inc. 1&2 - 20240318 - AE Review -- RETURNED RR	5/13/2024	5/13/2024
CDC-CCM-TIA-1310	Gap of 18 calendar days	5/13/2024	5/31/2024
CDC-CCM-TIA-1320	Submittal #095.8 - 08 4313 Aluminum Framed Storefronts Inc. 1&2 - 20240318 - AE Review -- AE APPROVED	5/31/2024	6/13/2024
CDC-CCM-TIA-1330	Submittal #095.8 - 08 4313 Aluminum Framed Storefronts Inc. 1&2 - 20240318 - Second Submission to DSA -- DSA APPROVED	6/13/2024	6/24/2024

These gaps account for 121 CDs. These gaps appear to represent instances in which Wickman failed to honor the "time is of the essence" requirements of the contract and, thus, are non-excusable delays. It is also reasonable to expect Wickman to complete the re-submittal process within one or two submissions; however, Wickman took at least eight re-submissions to achieve DSA approval. Therefore, any delays occurring after the first re-submissions are non-excusable.



This window represents 327 CDs of slippage, of which 288 CDs (Delay F, Exhibit C) are due to a lack of timely completion of re-submittals by Wickman. 39 CDs are due to the storefront re-design (Delay E, Exhibit C). The 288 CD delay is non-excusable and 39 CD delay is excusable but non-compensable due to concurrent delays.

Summary

Below is a table that summarizes the CPA. In short, had Wickman met and maintained the Contract requirements for a time extension, it would only be entitled to a 62 calendar day non-compensable time extension.

Table 5 - Summary of Windows Analysis

Window	Excusable, Non-compensable	Excusable, Compensable	Non-excusable
1	64	28	0
2	-8	-28	0
3	0	0	15
4	39	0	288
PREVIOUS TIME EXTENSION RECONCILITAITON	*-33*	*0*	*0*
TOTAL	***62***	***0***	***303***

OTHER TIAs

After determining that Wickman appeared to have waived entitlement on TIAs 05.3 and 09, Civic CM went back and evaluated TIAs 06, 07, 10, 14, and 18. Entitlement appears to have been waived on all these TIAs as well due to the timing of submission of Notice of Delay.

Civic CM evaluated the technical integrity of Wickman's TIAs 06, 07, 10, 14, and 18 to ensure compliance with the contract and consistency with published industry-recommended practices, such as FSA 2011. Civic CM identified that Wickman's selection of the pre-impact schedule used to host the fragnets modeling the alleged delay events for TIAs 07, 10, 14, and 18 was inconsistent with FSA 2011. FSA 2011 states, "Select the as-planned network to be utilized as the un-impacted schedule. If not using the baseline, **select the contemporaneous update that existed just prior to the initial delay that is to be evaluated."**

In other words, select the contemporaneous update that existed just before the delay began to drive the critical path. Wickman used the baseline schedule when they should have selected the schedule update prior to each TIA impact to model the delay instead. This is consistent with the findings from evaluation of TIAs 05.3 and 09. TIA-06 used the baseline and that practice was appropriate given the date of occurrence of the impact was February 3, 2023.

To model the impact for TIAs 07, 10, 14, and 18, Wickman selected the baseline schedule with a data date of January 4, 2023. The baseline *could* be utilized but would be inappropriate given that it is inconsistent with FSA 2011 recommended practices and that the baseline is several months earlier than the impact events for all these TIAs. The baseline schedule is three months prior to TIA-07, seven months prior to TIA-10, nine months prior to TIA-14, and eight months prior to TIA-18. Wickman also submitted multiple schedule updates before any of these issues allegedly began driving the critical path. Each schedule update would have more accurately shown the project's

contemporaneous condition just before the start of the delay event. Performing the TIA utilizing an older schedule can make it difficult to accurately represent the effect of the delay event because an earlier schedule does not capture the project's contemporaneous conditions just prior to the start of the delay event. The below table lists the associated Notice of Delay submissions along with some related comments, how many days Wickman was claiming, and whether entitlement was maintained for each item.

Table 6 - Summary of Other TIAs with Associated Notice of Delay

TIA	NOD	Comments	Days Claimed	Entitlement Maintained
TIA-06	NOD-02	RFI-016 submitted February 3, 2023. Notice of Delay 002 sent on March 14, 2023, 39 days after Wickman was aware of the delay. Entitlement not maintained	15	No
TIA-07	NOD-07	RFI-103 submitted April 25, 2023. RFI-103.1 submitted May 15, 2023. RFI-107 submitted May 9, 2023. PCO-008 submitted May 19, 2023. Notice of Delay 007 sent May 10, 2023, 15 days after Wickman was aware of the delay. Entitlement not maintained.	27	No
TIA-10	NOD-09	RFI-200 for Mechanical and Fire Protection Supports submitted August 17, 2023. Notice of Delay 009 sent on September 14, 2023, 28 days after Wickman was aware of the delay. Entitlement is not maintained.	134	No
TIA-14	None Submitted	RFI-284 10/25/2023; Response 11/7/2023; PCO-030 and PCO-031 11/21/2023; RFIs 242, 502, 507, 509 submitted between 12/26/2023 and 5/17/2024; There does not appear to be a Notice of Delay submitted for this delay. Entitlement not maintained.	134	No
TIA-18	NOD-13	RFI-254 was submitted September 29, 2023. Notice of Delay 013 sent on March 14, 2024, 167 days after Wickman was aware of the delay. Entitlement is not maintained.	6	No

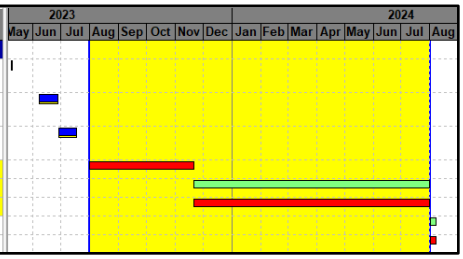
OTHER OUTSTANDING ISSUES

Civic CM was also asked to evaluate the delay with Wickman's execution of structural steel welding and subsequent re-work. Civic CM was provided with a timeline of events that ultimately affected the finish of the two brace frame repair activities, CDC1-3188 and CDC2-4008. The timeline of events provided to Civic CM assumes a finish of the structural steel delay and repair work of July 31, 2024. Civic CM utilized the July 2023 schedule update. To demonstrate the impact of the events from the timeline in the schedule, the forecast finish dates of CDC1-3188 and CDC2-4008 were changed to July 31, 2024. An activity was also added for Discovery of Brace Frame Issue to demonstrate how long it took to identify the needed repairs. To demonstrate the impact of the events from the timeline in the schedule, the forecast finish of CDC1-3188 and CDC2-4008 were changed to July 31, 2024. The total time from start of discovery (July 31, 2023), through completion of brace frame repair (July 31, 2024) is 365 days. This delay can also be viewed on the TIA Summary allowing the concurrency with TIA 09.4 to be seen. This would not

change the amount of time that Wickman would be due (231 CDs), had Wickman preserved its entitlement. Figure 5 below shows the referenced activities in the schedule.

Figure 5 – Structural Steel Delay Adjustments

#	Activity ID	Activity Name	OD	Start	Finish	TF
1	2023.07.Merritt_CDC_Schedule_Update					
2	CDC-2330	Steel Reinforcement, Embeds & Braceframe (S/S-740, 6/S/S-301, 15/S/S-721)	2	08-May-23 A	09-May-23 A	
3	CDC-2400	Structural Steel Frame - Columns, Braceframes, Girders & Beams (S/S-102)	15	07-Jun-23 A	27-Jun-23 A	
4	CDC-2405	Structural Steel Frame - Columns, Braceframes, Girders & Beams (L/S-102)	12	28-Jun-23 A	17-Jul-23 A	
5	CDC-2415	Disocvery of Brace Frame Issue	79	31-Jul-23	20-Nov-23	0
6	CDC1-3188	Brace Frame Repair by IBS (1st Floor)	171	21-Nov-23	31-Jul-24	46
7	CDC2-4008	Brace Frame Repair by IBS (2nd Floor)	171	21-Nov-23	31-Jul-24	0
8	CDC1-3189	Complete Interior Metal Framing at Brace Frames	5	01-Aug-24	07-Aug-24	46
9	CDC2-4009	Complete Interior Framing at Brace Frames	5	01-Aug-24	07-Aug-24	0



CLARIFICATIONS & RESERVATIONS

Please also note that the findings, observations, comments, or recommendations made within this work product are not intended to direct the contractor to revise their construction means and methods. This analysis is not intended to direct the contractor to complete any work above and beyond existing Contract requirements. This review and its attachments are a privileged and Confidential consultant work product protected by evidence code and are subject to revision. This review is subject to modification pending any additional information that may be produced, including, but not limited to, other schedule submissions, expert reports or responses from the contractor, and additional data that may become available after the date of this report. Should additional information be produced that may change the findings, observations, comments, or recommendations captured within this work product, Civic CM reserves the right to submit a supplemental report or addendum.

APPENDIX

- Exhibit A – TIA Summary
- Exhibit B – Civic CM Independently Prepared TIA
- Exhibit C – Delay Trend Chart